

Title

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Sources and Sample Selection

- Center for Medicare and Medicaid Services (CMS)
 - ▶ Nursing Home Compare (NHC) Archive
 - ▶ Healthcare Cost Report Information System (HCRIS)
 - ▶ Payroll Based Journal (PBJ)
- Facility-month-year level
- 48 Contiguous U.S. States over the period January 2017 – June 2024
- Freestanding facilities
 - ▶ Hospital based facilities make up 12% of all facilities in the U.S.

Identifying Ownership Changes - Background

- HCRIS ownership change flag
 - ▶ May reflect purely legal restructuring
- We validate using ownership data in the NHC archive
 - ▶ Reports every level of ownership, management through direct owners
 - ▶ Limited Liability Corporations
- We identify ownership changes at the highest level of ownership present
- Indirect $\implies$ Direct $\implies$ Partnership Interest
- ESTES JAMES (84%); BURCHFIELD JOHN (10%); LEE CLAUDE (6%) $\rightarrow$ JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%); JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)

Identifying Ownership Changes: Example (Family Continuity)

- Ownership records often reflect legal restructuring within the same individual or family
- **Example from the data:**
 - ▶ *Before:*
 - ★ ESTES JAMES (84%)
 - ★ BURCHFIELD JOHN (10%)
 - ★ LEE CLAUDE (6%)
 - ▶ *After:*
 - ★ JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
 - ★ JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)
- Despite the restructuring:
 - ▶ Control remains within the Estes family
 - ▶ No meaningful change in economic ownership
- We do **not** flag this as an ownership change

Why Ownership Is Non-Trivial: Example

Ownership Level	Entity Type	Owner Name	Share
<i>Direct ownership</i>			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
<i>Indirect ownership</i>			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

- Ownership is layered across direct and indirect entities
- Economic control is not obvious from direct ownership alone

Identifying Ownership Changes - Criteria

- ➊ Majority (50%) of ownership transfers from month $t - 1$ to month t
- ➋ Partial percentages we normalize and missing we assign equal weights across
- ➌ We don't flag ownership changes when 80% stake remains in the same family or for changes in legal names (e.g., a change from "Smith Holdings LLC" to "Smith Family Partners LP")
- ➍ Our sample then included facilities that facilities that:
 - ➊ Have no ownership changes in either dataset
 - ➋ Have exactly one change in each dataset within six months of each other
- ➎ For these facilities we define the change month as the first month the NHC archive indicates a change

Dependent Variables - Nurse Staffing Levels

- The staff types we look at are RN's, LPN's, CNA's, and total staffing
- PBJ
- Hours per patient day (HPPD) for each staff type s , facility i , and period t

$$\text{HPPD}_{i,t} = \frac{H_{s,i,t}}{R_{s,i,t}} \quad (1)$$

where $H_{s,i,t}$ is the total hours worked by the staff type s during month t for facility i and $R_{s,i,t}$ is the total resident days in that facility month

Dependent Variables - Nurse Staffing Levels

- We exclude a small number of observations to align with the implausible criteria outlined in the CMS NHC Technical Users Guide where:
 - ① RN and LPN HPPD are both 0 or
 - ② Total HPPD is below 1.5 or above 12 or
 - ③ CNA HPPD exceeds 5.25

Summary Statistics

- Manual fine tuning and cleaning
- Final sample:
 - ▶ 7,806 facilities
 - ▶ 632,231 facility-month observations

Data Sources and Unit of Observation

- **Nursing Home Compare (NHC) Archive**

- ▶ Monthly ownership roster with *every reported ownership layer* (managing org → direct owners → indirect owners)
- ▶ Used to **identify** and **date** ownership changes using ownership shares

- **Healthcare Cost Report Information System (HCRIS)**

- ▶ Annual cost report information + **ownership change flag**
- ▶ Used as an **initial flag** that a change may have occurred

- **Payroll Based Journal (PBJ)**

- ▶ Payroll-reported staffing hours by staff type
- ▶ Used to construct **hours per patient day (HPPD)** outcomes

- **Sample frame:** facility-month panel, 48 contiguous states, Jan 2017 – Jun 2024

Identifying Ownership Changes: Why Ownership Is Non-Trivial

- Ownership is often **layered** (organizations owning organizations), so economic control is not obvious from direct owners alone
- NHC reports ownership at multiple levels (managing org, direct, and indirect owners)

Ownership Level	Entity Type	Owner Name	Share
<i>Direct ownership</i>			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
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Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

Identifying Ownership Changes: Example (Legal Restructuring)

- HCRIS ownership change flags may reflect **purely legal restructuring** (not a real transfer of control)
- NHC ownership rosters allow us to verify whether economic control changes
- **Example from the data (same family control):**
 - ▶ *Before:* ESTES JAMES (84%); BURCHFIELD JOHN (10%); LEE CLAUDE (6%)
 - ▶ *After:* JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%); JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)
- Interpretation: restructuring into trusts/vehicles, but **control remains within the same family**
- We do **not** treat this as a meaningful ownership change

Identifying Ownership Changes: Definition and Sample

- **Step 1 (flag):** HCRIS ownership change flag identifies candidate change events
- **Step 2 (validate & date):** NHC ownership rosters identify whether control changes and when
- **Ownership change definition (NHC):**
 - ▶ An ownership change occurs between $t - 1$ and t when **majority control (50%)** transfers
 - ▶ Partial / missing shares are normalized; missing ownership shares are assigned equal weights within a level
 - ▶ We do not flag changes driven by name changes or within-family reclassification (e.g., trust/LLC relabeling)
- **Final sample restriction (agreement):**
 - ▶ Facilities with **(0,0)** changes in both datasets, or **(1,1)** with dates within 6 months
 - ▶ Change month is the first month NHC indicates a change

Dependent Variables: Nurse Staffing (PBJ)

- PBJ reports payroll-based staffing hours by staff category
- We study four staffing measures:
 - ▶ **Registered Nurses (RN)**: clinical assessment, medication administration, supervision
 - ▶ **Licensed Practical Nurses (LPN)**: clinical support, medication-related tasks (scope varies by state)
 - ▶ **Certified Nursing Assistants (CNA)**: direct daily care (bathing, feeding, mobility)
 - ▶ **Total nursing staff**: RN + LPN + CNA
- Outcome is **hours per patient day (HPPD)**: staffing intensity standardized by resident census

$$\text{HPPD}_{s,i,t} = \frac{H_{s,i,t}}{\text{ResidentDays}_{i,t}}$$

PBJ Cleaning and Final Sample

- We apply CMS technical guidance to remove implausible staffing reports:
 - ① RN HPPD = 0 **and** LPN HPPD = 0
 - ② Total HPPD < 1.5 or > 12
 - ③ CNA HPPD > 5.25
- Additional restrictions:
 - ▶ Drop facilities with beds < 15
 - ▶ Drop hospital-based facilities
- **Final sample:**
 - ▶ 7,806 facilities
 - ▶ 632,231 facility-month observations